

PHYS 486 - HW08

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1) Just like orbital angular momentum $\vec{L}$ and spin angular momentum $\vec{S}$, we can implement the same formalism for $\vec{J} = \vec{L} + \vec{S}$

For $j = \frac{3}{2}$, dimension of the subspace is $2j+1 = 4$. Basis of this Hilbert subspace are simult. eigenstates of J^2 and J_z operators which denoted as $|j m\rangle$

$$\begin{aligned} & \bullet |3/2, 3/2\rangle \Rightarrow [1 \ 0 \ 0 \ 0]^T \text{ in Hilbert subspace} \\ & \bullet |3/2, 1/2\rangle \Rightarrow [0 \ 1 \ 0 \ 0]^T \\ & \bullet |3/2, -1/2\rangle \Rightarrow [0 \ 0 \ 1 \ 0]^T \\ & \bullet |3/2, -3/2\rangle \Rightarrow [0 \ 0 \ 0 \ 1]^T \end{aligned} \quad \left. \vphantom{\begin{aligned} & \bullet |3/2, 3/2\rangle \Rightarrow [1 \ 0 \ 0 \ 0]^T \\ & \bullet |3/2, 1/2\rangle \Rightarrow [0 \ 1 \ 0 \ 0]^T \\ & \bullet |3/2, -1/2\rangle \Rightarrow [0 \ 0 \ 1 \ 0]^T \\ & \bullet |3/2, -3/2\rangle \Rightarrow [0 \ 0 \ 0 \ 1]^T \end{aligned}} \right\} \text{eigenbasis}$$

$$J^2 |j m\rangle = \hbar^2 j(j+1) |j m\rangle, \quad J_z |j m\rangle = \hbar m |j m\rangle$$

$$J^2 |3/2, m\rangle = \hbar^2 \frac{3}{2} \cdot \frac{5}{2} |3/2, m\rangle = \frac{15\hbar^2}{4} |3/2, m\rangle \Rightarrow$$

$$J^2 = \frac{15\hbar^2}{4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\hookrightarrow$ since basis are eigenstates, L^2 will be diagonal

$$J_z |3/2, m\rangle = m\hbar |3/2, m\rangle \Rightarrow J_z = \frac{\hbar}{2} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

* Now to find J_x and J_y we'll first determine $J_{\pm}$ (raising/lowering operators)

$$\text{Recall that } J_{\pm} |j m\rangle = \hbar \sqrt{j(j+1) - m(m\pm 1)} |j, m\pm 1\rangle$$

$$J_+ |3/2, 3/2\rangle = 0 \Rightarrow \text{raising max } m \text{ gives zero}$$

$$J_+ |3/2, 1/2\rangle = \hbar \sqrt{15/4 - 3/4} |3/2, 3/2\rangle = \hbar \sqrt{3} |3/2, 3/2\rangle$$

$$J_+ |3/2, -1/2\rangle = \hbar \sqrt{15/4 + 1/4} |3/2, 1/2\rangle = 2\hbar |3/2, 1/2\rangle$$

$$J_+ |3/2, -3/2\rangle = \hbar \sqrt{15/4 - 9/4} |3/2, -1/2\rangle = \hbar \sqrt{3} |3/2, -1/2\rangle$$

$$J_+ = \hbar \begin{bmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

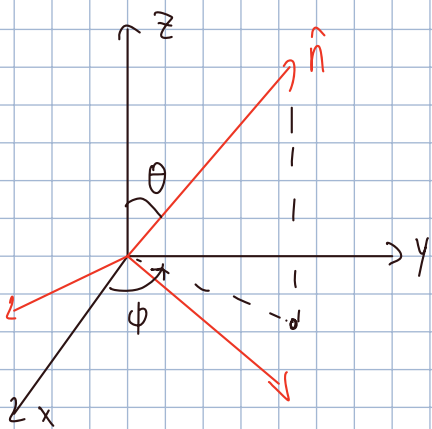
$$\left. \begin{aligned}
 J_- |3/2, 3/2\rangle &= \hbar \sqrt{15/4 - 3/4} |3/2, 1/2\rangle = \sqrt{3} \hbar |3/2, 1/2\rangle \\
 J_- |3/2, 1/2\rangle &= \hbar \sqrt{15/4 + 1/4} |3/2, -1/2\rangle = 2\hbar |3/2, -1/2\rangle \\
 J_- |3/2, -1/2\rangle &= \hbar \sqrt{15/4 - 3/4} |3/2, -3/2\rangle = \sqrt{3} \hbar |3/2, -3/2\rangle \\
 J_- |3/2, -3/2\rangle &= 0 \Rightarrow \text{lowering min m state yields zero}
 \end{aligned} \right\} J_- = \hbar \begin{bmatrix} 0 & 0 & 0 & 0 \\ \sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{bmatrix}$$

Now using $J_x = \frac{1}{2}(J_+ + J_-) = \frac{\hbar}{2} \begin{bmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{bmatrix}$

$$J_y = \frac{1}{2i}(J_+ - J_-) = \frac{\hbar}{2} \begin{bmatrix} 0 & -i\sqrt{3} & 0 & 0 \\ i\sqrt{3} & 0 & -2i & 0 \\ 0 & 2i & 0 & -i\sqrt{3} \\ 0 & 0 & i\sqrt{3} & 0 \end{bmatrix}$$

2) $\vec{L} \cdot \hat{n} = L_n = \sin\theta \cos\phi L_x + \sin\theta \sin\phi L_y + \cos\theta L_z$

2.a) We can physically define L_n as the operator L_z after coordinate system has been rotated so that z -axis points along direction $\hat{n}(\theta, \phi)$



This rotation of ϕ around z -axis, θ around y -axis is achieved by

$$U(\theta, \phi) = \exp\left(-\frac{i}{\hbar} \phi L_z\right) \exp\left(-\frac{i}{\hbar} \theta L_y\right) \Rightarrow U^\dagger = U_{\text{unitary}}$$

$\hookrightarrow$ Angular momentum operators are generators for rotation
 $\hookrightarrow$ See eq. 10 & 11 of Discussion 10

Then $L_n = U L_z U^\dagger \Rightarrow$ how operators transform under unitary U

$\rightarrow$ Let $|m\rangle$ be eigenvectors of L_z such that $L_z |m\rangle = m\hbar |m\rangle$

Let us define the rotated eigenstates as $|\psi'_m\rangle = U |m\rangle$ and apply L_n to these new states to find eigvals

$$L_n |\Psi'_m\rangle = U L_z U^\dagger = (U L_z U^\dagger) (U |m\rangle) = U L_z |m\rangle = U (m\hbar |m\rangle)$$

$$L_n |\Psi'_m\rangle = m\hbar U |m\rangle = m\hbar |\Psi'_m\rangle$$

↳ Eigenvalues of L_n are still $m \in \{-l, \dots, 0, \dots, l\}$
with eigenvectors $U |m\rangle$.

$$\star \text{ Eigenvalues of } L_n \text{ are } m \in \{-l, -l+1, \dots, -1, 0, 1, \dots, l-1, l\}$$

b)

$$\langle L_n \rangle = \sin\theta \cos\phi \langle L_x \rangle + \sin\theta \sin\phi \langle L_y \rangle + \cos\theta \langle L_z \rangle$$

$$\langle L_x \rangle = \langle \frac{1}{2} (L_+ + L_-) \rangle = \frac{1}{2} \langle L_+ \rangle + \frac{1}{2} \langle L_- \rangle = 0 \quad \text{since } L_{\pm} |l, m\rangle \text{ is orthogonal to } |l, m\rangle$$

$$\langle L_y \rangle = \langle \frac{1}{2i} (L_+ - L_-) \rangle = 0 \quad \text{similarly}$$

$$\langle L_z \rangle = \langle l, m | L_z | l, m \rangle = m\hbar \langle l, m | l, m \rangle = m\hbar$$

$$\langle L_n \rangle = \cos\theta \langle L_z \rangle = \boxed{m\hbar \cos\theta}$$

$$L_n^2 = \sin^2\theta \cos^2\phi L_x^2 + \sin^2\theta \sin^2\phi L_y^2 + \cos^2\theta L_z^2 + 2\sin^2\theta \cos\phi \sin\phi (L_x L_y + L_y L_x) + 2\sin\theta \cos\theta \cos\phi (L_x L_z + L_z L_x) + 2\sin\theta \cos\theta \sin\phi (L_y L_z + L_z L_y)$$

$$\bullet \quad L_y L_x + L_x L_y = \frac{1}{4i} [(L_+ - L_-)(L_+ + L_-) + (L_+ + L_-)(L_+ - L_-)] = \frac{1}{4i} (L_+^2 - L_-^2)$$

$$\langle L_y L_x + L_x L_y \rangle = \frac{1}{4i} (\langle L_+^2 \rangle - \langle L_-^2 \rangle) = 0 \quad \text{since } L_{\pm}^2 |l, m\rangle \perp |l, m\rangle$$

$$\bullet \quad \langle L_x L_z \rangle + \langle L_z L_x \rangle = m\hbar \langle L_x \rangle + \underbrace{m\hbar}_{(m\hbar)^* = m\hbar} \langle L_x \rangle = 0 \quad \text{since } L_z \text{ leaves } |l, m\rangle \text{ unchanged but } L_{\pm} \text{ in } L_x \text{ yields in inner prod. of orthogonal states.}$$

$$\bullet \quad \langle L_y L_z \rangle + \langle L_z L_y \rangle = 2m\hbar \langle L_y \rangle = 0 \quad \text{similarly.}$$

$$\text{Then we're left with } \langle L_n^2 \rangle = \sin^2\theta \cos^2\phi \langle L_x^2 \rangle + \sin^2\theta \sin^2\phi \langle L_y^2 \rangle + \cos^2\theta \langle L_z^2 \rangle$$

$$L^2 = L_x^2 + L_y^2 + L_z^2 \Rightarrow L_x^2 + L_y^2 = L^2 - L_z^2$$

* Recall that due to symmetry of $|m, l\rangle$ eigenstates around $L_z \Rightarrow \langle L_x^2 \rangle = \langle L_y^2 \rangle$
 $\hookrightarrow$ From previous homeworks.

$$\langle L_x^2 \rangle = \langle L_y^2 \rangle = \frac{\langle L^2 \rangle - \langle L_z^2 \rangle}{2} = \frac{\hbar^2 l(l+1) - (m\hbar)^2}{2} = \frac{\hbar^2}{2} [l(l+1) - m^2]$$

where we have used $L^2 |m, l\rangle = \hbar^2 l(l+1) |m, l\rangle$ and $L_z |m, l\rangle = m\hbar |m, l\rangle$

Then $\langle L_n^2 \rangle = (\sin^2\theta \cos^2\phi + \sin^2\theta \sin^2\phi) \frac{\hbar^2}{2} [l(l+1) - m^2] + m^2 \hbar^2 \cos^2\theta$

$$\langle L_n^2 \rangle = \frac{\hbar^2}{2} [l(l+1) - m^2] \sin^2\theta + m^2 \hbar^2 \cos^2\theta$$

3) $S_n = \vec{S} \cdot \hat{n} = \sin\theta \cos\phi S_x + \sin\theta \sin\phi S_y + \cos\theta S_z$ (from Question 2)

$$S_x = \frac{\hbar}{2} \sigma_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \sigma_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{\hbar}{2} \sigma_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

3.a)
$$S_n = \frac{\hbar}{2} (\sin\theta \cos\phi \sigma_x + \sin\theta \sin\phi \sigma_y + \cos\theta \sigma_z) = \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}$$

3.b)
$$\det(S_n - \lambda I) = \left(\frac{\hbar}{2} \cos\theta - \lambda \right) \left(-\frac{\hbar}{2} \cos\theta - \lambda \right) - \left(\frac{\hbar}{2} \sin\theta e^{-i\phi} \right) \left(\frac{\hbar}{2} \sin\theta e^{i\phi} \right) = 0$$

$$\left(\lambda - \frac{\hbar}{2} \cos\theta \right) \left(\lambda + \frac{\hbar}{2} \cos\theta \right) - \frac{\hbar^2}{4} \sin^2\theta = \lambda^2 - \frac{\hbar^2}{4} (\sin^2\theta + \cos^2\theta) = 0$$

$$\boxed{\lambda = \pm \frac{\hbar}{2}} \Rightarrow \text{eigenvalues of } L_n \text{ are } \pm \frac{\hbar}{2}$$

Let us now calculate the eigenvectors of L_n associated with $\pm \frac{\hbar}{2}$

$$\lambda = \pm \hbar/2$$

$$\text{Let } |\chi_+\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow \left(S_n - \frac{\hbar}{2} I\right) \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\frac{\hbar}{2} \begin{pmatrix} \cos\theta - 1 & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta - 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow (\cos\theta - 1)a + \sin\theta e^{-i\phi} b = 0$$

$$b = \frac{1 - \cos\theta}{\sin\theta} e^{i\phi} a$$

$$\text{Using } 1 - \cos\theta = 2 \sin^2 \theta/2, \quad \sin\theta = 2 \sin \theta/2 \cos \theta/2, \quad b = \frac{\sin \theta/2}{\cos \theta/2} e^{i\phi} a$$

$$|\chi_+\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) e^{i\phi} \end{pmatrix}$$

$$\lambda = \pm \hbar/2$$

$$\text{Let } |\chi_-\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow \left(S_n + \frac{\hbar}{2} I\right) \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\frac{\hbar}{2} \begin{pmatrix} \cos\theta + 1 & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta + 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow (\cos\theta + 1)a + \sin\theta e^{-i\phi} b = 0$$

$$b = -\frac{1 + \cos\theta}{\sin\theta} e^{i\phi} a$$

$$\text{Using } 1 + \cos\theta = 2 \cos^2 \theta/2, \quad \sin\theta = 2 \sin \theta/2 \cos \theta/2, \quad b = -\frac{\cos(\theta/2)}{\sin(\theta/2)} e^{i\phi} a$$

$$|\chi_-\rangle = \begin{pmatrix} \sin(\theta/2) \\ -\cos(\theta/2) e^{i\phi} \end{pmatrix}$$

→ Eigenvectors are $|\chi_+\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) e^{i\phi} \end{pmatrix}$ and $|\chi_-\rangle = \begin{pmatrix} \sin(\theta/2) \\ -\cos(\theta/2) e^{i\phi} \end{pmatrix}$ associated with $+\hbar/2$ & $-\hbar/2$ respectively.

$$\langle \chi_- | \chi_+ \rangle = \begin{bmatrix} \sin(\theta/2) & -\cos(\theta/2) e^{-i\phi} \end{bmatrix} \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2) e^{i\phi} \end{bmatrix}$$

$$= \sin(\theta/2) \cos(\theta/2) - \cos(\theta/2) \sin(\theta/2) e^{i\phi} e^{-i\phi} = 0 \Rightarrow \text{eigenvectors are orthogonal}$$

3.c) This device measures S_n . We want to find fraction of electrons that will be measured to have spin-up along $\hat{n}$. By measurement postulate, Prob. of measurement is given by:

$$P(\chi_+) = |\langle \chi_+ | \psi_{\text{initial}} \rangle|^2$$

→ $|\psi_{\text{initial}}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow$ spin-up in z -direction

$$P(\chi_+) = \left| \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2)e^{-i\phi} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2 = \boxed{\cos^2(\theta/2)}$$

→ For $\theta=0, \phi=0$: $P(\chi_+) = \cos^2 0 = 1$

↳ Measurement direction points along z -axis. Since particles are prepared in $+z$ state, we must measure them to be spin up 100% of the time ✓

→ For $\theta=\pi/2, \phi=0$: $P(\chi_+) = \cos^2(\pi/4) = (1/\sqrt{2})^2 = 1/2$

↳ Measurement direction points along x -axis. An electron with definite spin along z is an equal superposition of spin states along x .

$|+z\rangle = \frac{1}{\sqrt{2}}|+x\rangle + \frac{1}{\sqrt{2}}|-x\rangle$. Thus, a measurement along x -axis should yield spin-up half of the time and spin-down 50% of time. We recover expected result ✓